

NBER Behavioral Macroeconomics Research Boot Camp

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Lecture 1: Noisy information models and rational inattention

- *Gabaix, X. (2014). A sparsity-based model of bounded rationality. *The Quarterly Journal of Economics*
- *Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*,
- *Mackowiak, B., & Wiederholt, M. (2009). Optimal sticky prices under rational inattention. *American Economic Review*.
- *Gabaix, X. (2019). Behavioral inattention. *Handbook of Behavioral Economics: Applications and Foundations*.

Theoretical and psychological foundations:

- Sims, C. A. (2010). Rational inattention and monetary economics. In *Handbook of Monetary Economics*. Vol. 3.
- Anderson, J. R. (2014). Cognitive psychology and its implications (8th Edition). Chapter 3.
- Caplin, A., & Dean, M. (2015). Revealed preference, rational inattention, and costly information acquisition. *American Economic Review*.
- De Oliveira, H., Denti, T., Mihm, M., & Ozbek, K. (2017). Rationally inattentive preferences and hidden information costs. *Theoretical Economics*.
- Maćkowiak, B., Matějka, F., & Wiederholt, M. (2018). Dynamic rational inattention: Analytical results. *Journal of Economic Theory*. *Mimeo*.
- Caplin, A., Dean, M., & Leahy, J. (2019). Rationally inattentive behavior: Characterizing and generalizing Shannon entropy. *Mimeo*.
- Afrouzi, H., & Yang, C. (2020). Dynamic Rational Inattention and the Phillips Curve. *Mimeo*.
- Miao, J., Wu, J., & Young, E. (2020). Multivariate Rational Inattention. *Mimeo*.
- Hébert, B. M., & Woodford, M. (2020). Neighborhood-Based Information Costs. *Mimeo*.
- Pomatto, L., Strack, P., & Tamuz, O. (2020). The cost of information. *American Economic Review*.
- Woodford, M. (2019). Modeling Imprecision in Perception, Valuation, and Choice. *Annual Review of Economics*.
- Khaw, M. W., Li, Z., & Woodford, M. (2021). Cognitive imprecision and small-stakes risk aversion. *The Review of Economic Studies*.
- Frydman, C., & Jin, L. J. (2022). Efficient coding and risky choice. *Quarterly Journal of Economics*.
- Hébert, B., & Woodford, M. (2021). Neighborhood-based information costs. *American Economic Review*.

Applications:

- Van Nieuwerburgh, S., & Veldkamp, L. (2009). Information immobility and the home bias puzzle. *The Journal of Finance*.
- Van Nieuwerburgh, S., & Veldkamp, L. (2010). Information acquisition and under-diversification. *The Review of Economic Studies*.
- Maćkowiak, B., & Wiederholt, M. (2015). Business cycle dynamics under rational inattention. *The Review of Economic Studies*.
- Matějka, F., & McKay, A. (2015). Rational inattention to discrete choices: A new foundation for the multinomial logit model. *American Economic Review*.
- Kacperczyk, M., Van Nieuwerburgh, S., & Veldkamp, L. (2016). A rational theory of mutual funds' attention allocation. *Econometrica*.
- Matějka, F. (2016). Rationally inattentive seller: Sales and discrete pricing. *The Review of Economic Studies*.

Lecture 2. Survey evidence on expectations formation

- *Coibion, O., & Gorodnichenko, Y. (2012). What can survey forecasts tell us about information rigidities?. *Journal of Political Economy*.
- *Coibion, O., & Gorodnichenko, Y. (2015). Information rigidity and the expectations formation process: A simple framework and new facts. *American Economic Review*.
- *Bordalo, P., Gennaioli, N., Ma, Y., & Shleifer, A. (2020). Over-reaction in macroeconomic expectations. *American Economic Review*.
- *Angeletos, G. M., Huo, Z., & Sastry, K. A. (2020). Imperfect macroeconomic expectations: Evidence and Theory. *NBER Macroeconomics Annual*.

Theory:

- Evans, G. W., & Honkapohja, S. (2001). *Learning and expectations in macroeconomics*. Princeton University Press.
- Mankiw, N. G., & Reis, R. (2002). Sticky information versus sticky prices: a proposal to replace the New Keynesian Phillips curve. *The Quarterly Journal of Economics*.
- Reis, R. (2006). Inattentive producers. *The Review of Economic Studies*.
- Reis, R. (2006). Inattentive consumers. *Journal of Monetary Economics*.
- Alvarez, F. E., Lippi, F., & Paciello, L. (2016). Monetary shocks in models with inattentive producers. *The Review of Economic Studies*.
- Fuster, A., Laibson, D., & Mendel, B. (2010). Natural expectations and macroeconomic fluctuations. *Journal of Economic Perspectives*, 24(4), 67-84.
- Fuster, A., Hebert, B., & Laibson, D. (2012). Natural expectations, macroeconomic dynamics, and asset pricing. *NBER Macroeconomics Annual*.
- Bordalo, P., Gennaioli, N., & Shleifer, A. (2018). Diagnostic expectations and credit cycles. *The Journal of Finance*.

- Stevens, L. (2020). Coarse pricing policies. *The Review of Economic Studies*.
- Da Silva, R. A., Sung, Y., & Woodford, M. (2021). Optimally Imprecise Memory and Biased Forecasts. *NBER Working Paper*.

Evidence:

- Mankiw, N. G., Reis, R., & Wolfers, J. (2003). Disagreement about inflation expectations. *NBER Macroeconomics Annual*.
- Chetty, R., Looney, A., & Kroft, K. (2009). Salience and taxation: Theory and evidence. *American Economic Review*.
- Greenwood, R., & Shleifer, A. (2014). Expectations of returns and expected returns. *The Review of Financial Studies*
- Gennaioli, N., Ma, Y., & Shleifer, A. (2016). Expectations and investment. *NBER Macroeconomics Annual*.
- Khaw, M. W., Stevens, L., & Woodford, M. (2017). Discrete adjustment to a changing environment: Experimental evidence. *Journal of Monetary Economics*.
- Coibion, O., Gorodnichenko, Y., & Kumar, S. (2018). How do firms form their expectations? new survey evidence. *American Economic Review*.
- Afrouzi, H., Kwon, S. Y., Landier, A., Ma, Y., & Thesmar, D. (2021). Overreaction and Working Memory. *NBER Working Paper*.
- Malmendier, U., & Nagel, S. (2011). Depression babies: do macroeconomic experiences affect risk taking?. *The Quarterly Journal of Economics*.
- Malmendier, U., & Nagel, S. (2016). Learning from inflation experiences. *The Quarterly Journal of Economics*.
- Andrade, P., Coibion, O., Gautier, E., & Gorodnichenko, Y. (2020). No Firm is an Island? How Industry Conditions Shape Firms' Aggregate Expectations. *Journal of Monetary Economics*.

Lecture 3. Dampening general equilibrium: Models of imperfect coordination

- *Nagel, R. (1995). Unraveling in guessing games: An experimental study. *The American Economic Review*.
- *Morris, S., & Shin, H. S. (2002). Social value of public information. *American Economic review*.
- *Camerer, C. F., Ho, T. H., & Chong, J. K. (2004). A cognitive hierarchy model of games. *The Quarterly Journal of Economics*.
- *Angeletos, G. M., & Lian, C. (2022). Dampening General Equilibrium: Incomplete Information and Bounded Rationality. *Handbook of Economic Expectations*.

Noisy (incomplete) information as a model of imperfect coordination:

- Angeletos, G. M., & Pavan, A. (2007). Efficient use of information and social value of information. *Econometrica*.
- Angeletos, G-M., & Lian, C. (2016). "Incomplete information in

macroeconomics: Accommodating frictions in coordination." *Handbook of Macroeconomics*. Vol. 2. Chapter 7.

Level-k thinking:

- Stahl II, D. O., & Wilson, P. W. (1994). Experimental evidence on players' models of other players. *Journal of Economic Behavior & Organization*.
- Stahl, D. O., & Wilson, P. W. (1995). On players' models of other players: Theory and experimental evidence. *Games and Economic Behavior*.
- Ho, T. H., Camerer, C., & Weigelt, K. (1998). Iterated dominance and iterated best response in experimental "p-beauty contests". *American Economic Review*.
- Bosch-Domenech, A., Montalvo, J. G., Nagel, R., & Satorra, A. (2002). One, two, (three), infinity,...: Newspaper and lab beauty-contest experiments. *American Economic Review*.
- Costa-Gomes, M. A., & Crawford, V. P. (2006). Cognition and behavior in two-person guessing games: An experimental study. *American Economic Review*, 96(5), 1737-1768.
- Crawford, V. P., Costa-Gomes, M. A., & Iriberri, N. (2013). Structural models of nonequilibrium strategic thinking: Theory, evidence, and applications. *Journal of Economic Literature*.
- Coibion, O., Gorodnichenko, Y., Kumar, S., & Ryngaert, J. (2021). Do you know that i know that you know...? higher-order beliefs in survey data. *Quarterly Journal of Economics*.

Cursed equilibrium:

- Grossman, S. J., & Stiglitz, J. E. (1980). On the impossibility of informationally efficient markets. *The American Economic Review*.
- Eyster, E., & Rabin, M. (2005). Cursed equilibrium. *Econometrica*.
- Eyster, E., Rabin, M., & Vayanos, D. (2019). Financial markets where traders neglect the informational content of prices. *The Journal of Finance*.

Lecture 4. Macroeconomics with imperfect coordination

- *Woodford, M. (2003). Imperfect common knowledge and the effects of monetary policy. *Knowledge, Information, and Expectations in Modern Macroeconomics: In Honor of Edmund S. Phelps*
- *Angeletos, G. M., & Lian, C. (2018). Forward guidance without common knowledge. *American Economic Review*
- *Farhi, E., & Werning, I. (2019). Monetary policy, bounded rationality, and incomplete markets. *American Economic Review*.
- *García-Schmidt, M., & Woodford, M. (2019). Are low interest rates deflationary? A paradox of perfect-foresight analysis. *American Economic Review*
- *Gabaix, X. (2020). A Behavioral New Keynesian model. *American Economic Review*.

- *Angeletos, G. M., & Lian, C. (2022). Dampening General Equilibrium: Incomplete Information and Bounded Rationality. *Handbook of Economic Expectations*.

Complementary approaches and applications:

- Nimark, Kristoffer (2008), "Dynamic Pricing and Imperfect Common Knowledge," *Journal of Monetary Economics*.
- Angeletos, G. M., & La'O, J. (2010). Noisy business cycles. *NBER Macroeconomics Annual*.
- Allen, Franklyn, Stephen Morris, and Hyun Song Shin (2006), "Beauty Contests and Iterated Expectations in Asset Markets," *Review of Financial Studies*.
- Woodford, M. (2013). Macroeconomic analysis without the rational expectations hypothesis. *Annual Review Economics*.
- Melosi, L. (2014). Estimating models with dispersed information. *American Economic Journal: Macroeconomics*.
- Angeletos, G-M., & Lian, C. (2016). "Incomplete information in macroeconomics: Accommodating frictions in coordination." *Handbook of Macroeconomics*. Vol. 2. Chapter 8.
- Kiley, M. T. (2016). Policy paradoxes in the New Keynesian model. *Review of Economic Dynamics*.
- Andrade, P., Gabbalo, G., Mengus, E., & Mojon, B. (2019). Forward guidance and heterogeneous beliefs. *American Economic Journal: Macroeconomics*.
- Carroll, C., Crawley, E., Slacalek, J., Tokunaka, K., & White, M. N. (2020). Sticky expectations and consumption dynamics. *American Economic Journal: Macroeconomics*.
- Woodford, M. (2019). Monetary policy analysis when planning horizons are finite. *NBER Macroeconomics Annual*.
- Iovino, L., & Sergeyev, D. (2020). Central bank balance sheet policies without rational expectations. *Mimeo*.
- Angeletos, G. M., & Huo, Z. (2021). Myopia and anchoring. *American Economic Review*.
- Woodford, M., & Xie, Y. (2021). Fiscal and Monetary Stabilization Policy at the Zero Lower Bound: Consequences of Limited Foresight. *Journal of Monetary Economics*.
- Angeletos, G. M., & Sastry, K. A. (2021). Managing Expectations: Instruments Versus Targets. *The Quarterly Journal of Economics*.
- Lian, C. (2021). A theory of narrow thinking. *Review of Economic Studies*.
- Gorodnichenko, Y., & Sergeyev, D. (2021). Zero lower bound on inflation expectations. *NBER working paper*.
- Angeletos, G. M., & Lian, C. (2022). Determinacy without the Taylor Principle. *NBER working paper*.
- Vimercati, R. B., Eichenbaum, M. S., & Guerreiro, J. (2022). Fiscal policy at the zero lower bound without rational expectations. *NBER working paper*.

Lecture 5. Confidence, sentiments, and animal spirits

- *Barro, R. J., & King, R. G. (1984). Time-separable preferences and intertemporal-substitution models of business cycles. *Quarterly Journal of Economics*.
- *Lorenzoni, G. (2009). A theory of demand shocks. *American Economic Review*.
- *Angeletos, G. M., & La'O, J. (2013). Sentiments. *Econometrica*.
- *Angeletos, G. M., Collard, F., & Dellas, H. (2020). Business Cycle Anatomy. *American Economic Review*.
- *Angeletos, G. M., & Lian, C. (2022). Confidence and the Propagation of Demand Shocks. *Review of Economic Studies*.

Notions:

- Pigou, A. C. (1927). *Industrial fluctuations*. Routledge.
- Kindleberger, C. P. (1978). *Manias, panics and crashes: a history of financial crises*. Palgrave Macmillan.
- Minsky, H. P. (1986). *Stabilizing an unstable economy*. New York: McGraw-Hill.
- Cochrane, J. H. (1994). Shocks. *Carnegie-Rochester Conference Series on Public Policy*.
- Akerlof, G. A., & Shiller, R. J. (2010). *Animal spirits: How human psychology drives the economy, and why it matters for global capitalism*.
- Shiller, R. J. (2017). Narrative economics. *American Economic Review*.

Evidence on the macro impact of news and noise shocks:

- Beaudry, P., & Portier, F. (2006). Stock prices, news, and economic fluctuations. *American Economic Review*.
- Barsky, Robert, and Eric Sims (2011), News Shocks and Business Cycles. *Journal of Monetary Economics*.
- Alexopoulos, M. (2011). Read all about it!! What happens following a technology shock?. *American Economic Review*.
- Blanchard, O. J., L'Huillier, J. P., & Lorenzoni, G. (2013). News, noise, and fluctuations: An empirical exploration. *American Economic Review*.
- Barsky, R. B., & Sims, E. R. (2012). Information, animal spirits, and the meaning of innovations in consumer confidence. *American Economic Review*.
- Barsky, Robert, Susanto Basu, and Keyoung Lee (2014), "Whither News Shocks?" *NBER Macro Annual*.
- Chahrour, R., & Jurado, K. (2018). News or noise? The missing link. *American Economic Review*.

Traditional multiple equilibria approaches:

- Blanchard, O. J., & Kahn, C. M. (1980). The solution of linear difference models under rational expectations. *Econometrica*.

- Diamond, P. A. (1982). Aggregate demand management in search equilibrium. *Journal of Political Economy*.
- Cooper, Russell, and Andrew John (1988), "Coordinating Coordination Failures in Keynesian Models," *Quarterly Journal of Economics*.
- Howitt, P., & McAfee, R. P. (1992). Animal spirits. *The American Economic Review*.
- Farmer, R. E., & Benhabib, J. (1994). Indeterminacy and increasing returns. *Journal of Economic Theory*.
- Farmer, R. E., & Guo, J. T. (1994). Real business cycles and the animal spirits hypothesis. *Journal of Economic Theory*.
- Obstfeld, M. (1996). Models of currency crises with self-fulfilling features. *European Economic Review*.
- Calvo, G. A. (1998). Capital flows and capital-market crises: the simple economics of sudden stops. *Journal of applied Economics*.
- Benhabib, J., & Farmer, R. E. (1999). Indeterminacy and sunspots in macroeconomics. *Handbook of macroeconomics*.
- Benhabib, J., & Wen, Y. (2004). Indeterminacy, aggregate demand, and the real business cycle. *Journal of Monetary Economics*.
- Benhabib, J., Wang, P., & Wen, Y. (2015). Sentiments and aggregate demand fluctuations. *Econometrica*.
- Kaplan, G., & Menzio, G. (2016). Shopping externalities and self-fulfilling unemployment fluctuations. *Journal of Political Economy*.

Models of news and noise shocks:

- Beaudry, P., & Portier, F. (2004). An exploration into Pigou's theory of cycles. *Journal of Monetary Economics*.
- Jaimovich, Nir, and Sergio Rebelo (2009), "Can News About the Future Drive the Business Cycle?," *American Economic Review*.
- Christiano, L., Ilut, C. L., Motto, R., & Rostagno, M. (2010). Monetary policy and stock market booms. *Macroeconomic Challenges: the Decade Ahead, Federal Reserve Bank of Kansas City, Policy Symposium, Jackson Hole Wyoming*.
- Schmitt-Grohé, S., & Uribe, M. (2012). What's news in business cycles. *Econometrica*.

Models of sentiments and animal spirits:

- Angeletos, G. M., Collard, F., & Dellas, H. (2018). Quantifying confidence. *Econometrica*.
- Angeletos, G. M. (2018). Frictional coordination. *Journal of the European Economic Association*.
- Bhandari, A., Borovička, J., & Ho, P. (2019). Survey data and subjective beliefs in business cycle models. *Mimeo*.
- Bordalo, P., Gennaioli, N., Shleifer, A., & Terry, S. J. (2020). Real credit cycles. *Mimeo*.

- Krishnamurthy, A., & Li, W. (2020). Dissecting mechanisms of financial crises: Intermediation and sentiment. *NBER working paper*.
- Beaudry, P., Galizia, D., & Portier, F. (2020). Putting the cycle back into business cycle analysis. *American Economic Review*.